

Suricata First-Run Startup Issue - Fixed

Problem

Symptom: Suricata fails to start on first run but works on second attempt.

```
First run:  x Failed to start Suricata
Second run:  △ Suricata already running
```

Root Cause

Race Condition in Startup Check

When Suricata starts in daemon mode (**-D**), the process:

1. Forks to background immediately
2. Initializes (loads rules, opens network interface) - **Takes 5-10 seconds**
3. If initialization fails, exits silently

The script was checking if Suricata was running after only **3 seconds**:

```
sleep 3 # Too short!
if pgrep -x suricata > /dev/null; then
    echo "Success"
else
    echo "Failed" # ← Premature failure
fi
```

Timeline (First Run):

```
0s: Script starts Suricata with -D
0s: Suricata forks to background
1s: Suricata loading rules...
2s: Suricata initializing interface...
3s: Script checks - process exists ✓
3s: Script exits
4s: Suricata still initializing...
5s: Suricata encounters error (rules not fully loaded)
6s: Suricata crashes/exits
7s: BUT process briefly existed, so script reported "success"
```

Why Second Run Works:

- First run's Suricata process might still be lingering

- OR Suricata left a PID file behind
- Script detects it and says "already running"
- Actually starts properly the second time

The Fix

Implemented Robust Startup Check with Retry Logic

```
# 1. Increased initial wait time
sleep 5 # Give Suricata more time to fork properly

# 2. Retry loop - wait up to 10 more seconds
RETRY_COUNT=0
MAX_RETRIES=5
while [ $RETRY_COUNT -lt $MAX_RETRIES ]; do
    if pgrep -x suricata > /dev/null; then
        # Found process, wait 1 more second to verify stability
        sleep 1
        if pgrep -x suricata > /dev/null; then
            # Still running - startup successful
            break
        fi
    fi
    RETRY_COUNT=$((RETRY_COUNT + 1))
    if [ $RETRY_COUNT -lt $MAX_RETRIES ]; then
        echo "Waiting for Suricata... ($RETRY_COUNT/$MAX_RETRIES)"
        sleep 2
    fi
done

# 3. Final check
if pgrep -x suricata > /dev/null; then
    echo "✓ Success"
else
    echo "✗ Failed"
    # Show actual error from logs
fi
```

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Benefits:

- ✓ Waits up to 15 seconds total (5 initial + 10 retry)
- ✓ Checks process stability (verifies it doesn't immediately crash)
- ✓ Provides progress feedback during waiting
- ✓ Works reliably on first run

Testing the Fix

```
# Stop everything
sudo pkill -f suricata
sudo pkill -f kafka

# Clean start - should work first time now
sudo ./run_afpacket_mode.sh
# Select option 1

# Expected output:
== Starting Suricata (AF_PACKET Mode) ==
Starting Suricata in AF_PACKET mode...
Waiting for Suricata to initialize...
✓ Suricata started successfully!
PID: 12345
```

Why 3 Seconds Wasn't Enough

Suricata Initialization Steps (can take 5-10 seconds):

1. Parse configuration file (1-2s)
2. Load detection rules (2-4s) ← **Slow on first run**
3. Initialize threads (1s)
4. Open network interface (1s)
5. Set up promiscuous mode (1s)
6. Start capture (1s)

First run is slower because:

- Rules compiled for first time
- Interface brought up from cold state
- Kernel buffers allocated
- No cached data

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Additional Improvements Made

1. **Startup feedback:** Shows "Waiting for Suricata to initialize..." message
2. **Retry counter:** Shows progress if taking longer than expected
3. **Stability check:** Waits 1 second and re-checks to ensure process didn't immediately crash
4. **Better error reporting:** If fails, points to actual log file

Files Modified

- `dpkg_suricata_ml_pipeline/scripts/03_start_suricata_afpacket.sh`
 - Changed: `sleep 3` → `sleep 5` + retry loop
 - Added: Stability check and progress feedback

Summary

Before:

- Checked too early (3 seconds)
- No retry logic
- Failed on first run ×

After:

- Waits appropriately (5 seconds + retries up to 15 seconds total)
- Verifies process stability
- Works reliably on first run ✓

Now run **sudo ./run_afpacket_mode.sh** and it should work perfectly the first time!